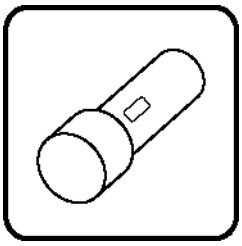
Troubleshooting



Machine Trouble

Problem	Cause	Correction
Safety button will not reset	Hydraulic oil overheated	Check temperature of water modulating valve
	Safety door open	Check safety doors
Machine alarms out before parison drops	Timer timed out (screw overfill)	Adjust timer. Adjust cycle times and/or rpm of drive
	Proximity overfill switch activated	Adjust cycle time and/or rpm of drive
Machine alarms out when molds should close after parison drop (stripper safety)	Limit switch is not actuated in the Up position	Check actuator cam
	Plastic hung up on stripper plate	Clean stripper plate
Machine alarms out as molds close (low pressure close safety)	Hydraulic low pressure set too low	Adjust low pressure higher
	Mechanical drag on tie bars	Lubricate tie bars
Machine alarms out as molds reach full open position	Machine out of resin	Refill hopper with pellets or regrind
Stripper plate does not go down	Solenoid coil burned out or valve spool does not move	Repair or replace parts
Stripper plate does not return to Up position	Valve spool does not move	Clean or replace valve spool
Temperature controller does not turn on heat zones	Open thermocouple	Replace thermocouple
	Defective controller	Replace controller
Temperature controller does not reach setpoint	Thermocouple short- circuited	Replace thermocouple
	Cooling fan relay stuck open	Replace cooling fan relay
	Temperature control module defective	Replace control card
Hydraulic pump motor does not start	Overload switch has tripped open at either: • Hydraulic motor • Gear box coolant motor • Drive motor	Reset overload

Machine Trouble

Problem	Cause	Correction
Extruder motor does not start	Overload switch has tripped open at either: • Hydraulic motor • Gear box coolant motor • Drive motor	Reset overload
Screw does not retract all or part of the way	A collar has formed on screw	Pour handle slugs into material hopper
	Lack of resin	Refill hopper with pellets or regrind
	Defective manual Charge button	Replace Charge button
	Burned out solenoid coil on Charge valve	Replace coil
Machine does not go into Auto Cycle	Defective Cycle Start button	Replace defective button
	Door air valves do not activate or pressure switch does not actuate	Adjust or replace valve and/or pressure switch
Molds do not close	Preblow, blow, and exhaust timers do not reset at end of cycle	Check setpoints on Timers screen; reset as necessary.
	Mold close solenoid energizes but valve does not shift	Replace solenoid
	Low pressure (hydraulic) set too low	Adjust low pressure
Some blowpins do not blow	Plugged blowpin	Clean blowpin

Bottle/Parison Formation Trouble

Problem	Cause	Correction
Air bubbles-holes in parison (fish eyes)	Moisture in resin	Check for moisture in virgin resin or regrind; replace if necessary
	Contamination	Check regrind and virgin resin supply for foreign material or bubbles
	Fill pressure too low	Adjust fill pressure valve

Bottle/Parison Formation Trouble

Problem	Cause	Correction
Air trapped	Stock temperature high	Change extruder temperature profile to lower stock temperature
	Mold and cavity surfaces dirty	Clean mold vents
	Mold temperature too high	Decrease mold temperature
	Blow air pressure too low	Increase blow air pressure
	Sand blasting worn off	Sand blast cavity
Bottles not blowing	Bottle weight incorrect	Adjust parison
	Parison support air too low	Increase support air
	Low blow air pressure	Increase blow air pressure
	Extruder surging	Check drive
	Container distribution incorrect	Adjust die
	Contamination	Dump and clean resin. Remove and clean die head tooling
Bottles sticking in mold	Bottle sidewalls incorrect	Adjust die
	Mold cavity dirty	Clean cavity
	Improper amount of material in flash area around ejector	Check thickness in ejector area, add/subtract as necessary
	Mold surface wet	Find and eliminate source of moisture
	Bottle too hot	Check chilled water temperature and flow rate Check air pressure Check for plugged needle

Bottle/Parison Formation Trouble

Problem	Cause	Correction
Bottle weight varying, extruder output surging	Extruder drawer magnet bridged or dirty (load on extruder load meter has dropped)	Clean magnet
	Feed throat of extruder bridged (load on extruder load meter has dropped)	Increase flow of cooling water to feed throat Reduce feed section barrel temperature
	Closed loop system low on water	Refill with distilled water
	Barrel zone under temperature	Check breaker on barrel heater Check barrel heater; replace if necessary
	Improper extruder profile	Adjust extruder temperature profile
Die lines	Dirty mandrel and die	Probe mandrel and die
	Die temperature too low	Increase die temperature
	Die head temperature too low	Increase die head temperature
	Resin temperature too low	Increase barrel temperature
Drop test failure	Bottle weight too low	Adjust parison
	Container wall distribution incorrect	Adjust die
	Not enough material in pinch-off area	Adjust weight of material in pinch-off area
	Pinch-off too sharp (new molds)	Stone pinch edge slightly
	Mold temperature too high	Decrease mold temperature
	Pinch-off dull (rolled over)	Send out mold for refurbishment
	Streak	Clean applicable tooling
Mismatched part lines	Alignment dowels worn	Replace alignment dowels
	Mold alignment poor	Realign mold (see mold set up)
Orange peel finish	Mold surface wet	Locate source of moisture and eliminate
Parison sag	Stock temperature high	Reduce temperature
	Parison hang time too long	Start mold close sequence earlier

Bottle/Parison Formation Trouble

Problem	Cause	Correction
Parison hooking	Die adjustment not concentric	Adjust die
	Pressure ring not concentric	Center the pressure ring
	Die head temperature not even	Check for head heater failure; replace heater Make sure all die head thermocouples are in contact with the die head
Trapped air	Mold and cavity surfaces dirty	Clean mold vents
	Mold temperature too high	Decrease mold temperature
	Blow air pressure too low	Increase blow air pressure
	Sand blasting worn off	Sand blast cavity